XAMS optical simulation: model, validation and first results

Geant4 optical simulation of the XAMS TPC with the meshes as built — geometry v4 from the drawings (TPC in the PTFE bucket and the cryostat vessel, PTFE PMT holders) · 9 October 2026 · optical_sim/ (local git), user guide: optical_sim/docs (MkDocs)

Summary.

• Question: is the low S2 light on the top PMT a geometric effect? **Partly.** Geometry and optics give an S2 top fraction of 0.52–0.58 over all plausible optical parameters and liquid levels. The data without a source come close (0.50–0.68 depending on S2 size, 0.55 at 10–20 kPE); the Cs-137 and Na-22 values (0.27–0.40) are far below anything geometric — they are the load effect of the top PMT, not geometry.

• **The geometry is right:** the top-sector light pattern versus position (asymmetries u, v) lies inside the narrow simulation band over ±24 mm, identically for every source.

• **Fit:** the shapes of S2 versus r, S1 versus depth and the sector pattern are reproduced. No-source data: ε_{S2} = 1.17, ε_{S1} = 0.74 on the top/bottom ratio; **Cs-137 photopeak (large signals): the same best optics, no central dip, and one factor ε_{S2} = 0.31 ≈ ε_{S1} = 0.33** (the top-PMT load). The no-source $\varepsilon_{S1} < \varepsilon_{S2}$ is the top small-signal loss: amstrax sums only S2/S1 samples ≥ 15 ADC (a single top photon keeps ~55 %). Low steel reflectivity is preferred; the other optical parameters are not constrained.

• **Central dip** of the S2 area fraction top (no source, small S2s): not optics, not a dead PMT cross, not the readout threshold (toy: < 1 % at S2 ≥ 3 kPE). It comes from an extra population of small S2s piled up at the reconstructed centre (density ×10–14 for 3–20 kPE, ×1.2 above 50 kPE) with low area fraction top, also in raw u–v. Their S1 does not fit their drift time (S1 AFT says 15 µs deeper, S2/S1 ≈ 13): accidental S1–S2 pairs, not optics.

• **Lower half (y < 0):** in an arc at r ≈ 10–18 mm the S2 top/bottom ratio is ~0.6–0.8 × the model, but only for interactions in the upper half of the drift region (drift < 25 µs). Mostly a population mixture: near-gate events (~2 mm below the gate, S2/S1 ≈ 600–1000) have a lower S2 AFT everywhere and form a larger share at y < 0, where bulk background events are rare; normal Cs-137 events there agree with the model. Top-channel gains are fine.

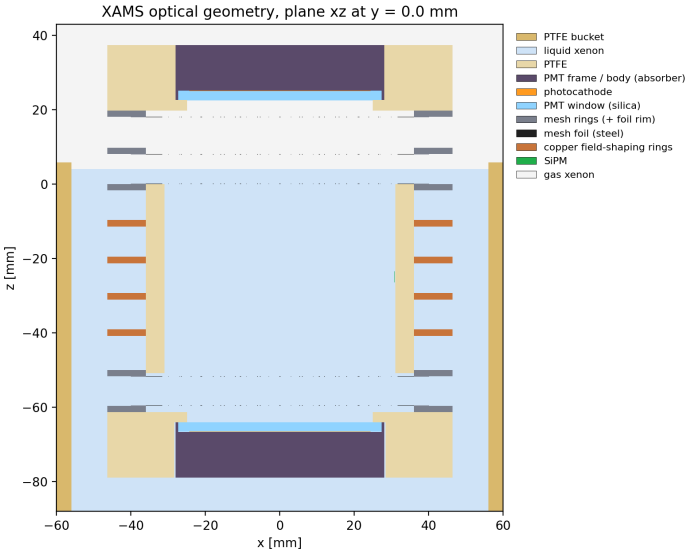
• Only ~25 % of the S2 photons reach a photocathode (QE 0.33 → 8.1 % detected); the mesh foils take ~40 %, the mesh and copper rings ~15 %, the bucket and the cryostat vessel ~12 %. The liquid level (2 → 4 mm) hardly matters. The surroundings (bucket, vessel) change nothing; the PTFE PMT holders add 7–9 % light on both PMTs.

• Design study: PTFE cylinders also in the gate–anode and anode–top-screen gaps raise the S2 light collection by +36 %, with the cathode–bottom-screen gap +43 % (S1 +28 %) — a factor ~1.4, not more.

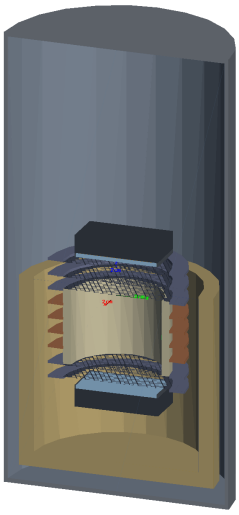
1. Model

One event = one 178 nm photon, started uniformly in the source volume with a random direction and polarisation and followed until it is detected or lost (G4OpticalPhysics: absorption, Rayleigh scattering, boundaries; no scintillation). S2: liquid surface to just below the anode foil, r < 30 mm; S1: cathode to gate, r < 30 mm. Every photon is recorded (origin, end point, how it ended, sensor, reflections); maps are made by binning the origins.

part	model (geometry v4, from the drawings)
meshes (5)	solid 0.15 mm steel foils (Ø80) with 456 square holes 2.45 mm (pitch 2.6 mm) etched in a staircase-shaped area (gate drawing; outer hole corners at r = 32.8 mm), the rest solid; all at 29.4° to the anodes. z: top screen +18, anode +8, gate 0, cathode −51.8, bottom screen −59.6 mm
mesh rings	steel, Ø72–92.6 mm, 1.8 mm assembly; the foil (Ø80) on one face, the ring body on the other side: below the gate and bottom screen, above the cathode, anode and top screen. The gate–anode gap is therefore open sideways between r = 36 and 46 mm
PTFE cylinder	Ø62/72 mm, 50.85 mm, in the bores of the gate and cathode rings (0.8 mm gap to the cathode foil); R 0.90–0.99, specular fraction 0–0.2, rest Lambertian
copper rings	four field-shaping rings Ø72–92.6 × 1.8 mm around the PTFE (8 mm gaps), R 0.2–0.4
xenon	liquid n = 1.69 up to the level (2 / 3 / 4 mm above the gate), Rayleigh 30–60 cm, absorption 1–10 m; gas n = 1.00 above
PMTs	two R12699: silica window 54.6 mm square, 2.5 mm, n = 1.60; photocathode 48.5 mm square with QE 0.33 (top: 4 sectors = ch1–ch4); frame and body absorbing
SiPMs	two, on the PTFE inner wall at z = −25 mm, azimuth 0° / 90°
PMT holders	PTFE plate between screen ring and PMT window (50 mm square opening) and block around each PMT (slot 56 / 56.9 mm), out to r = 46.3 mm
bucket, vessel	the TPC hangs in a PTFE bucket (Ø112/132 mm, 129 mm; the weir 1.78 mm below the rim sets the liquid level), inside the closed stainless-steel inner vessel (Ø146.9/152.4 mm, 305 mm); gas around the bucket. Bucket bottom, vessel ends and the gap below the bucket are defaults



Cross section at y = 0 from Geant4's own navigator (exactly what the simulation uses).



3-D view, cut open: PTFE bucket, PTFE (cream), mesh rings (grey), copper rings, PMTs, inside the vessel (without the PMT holders).

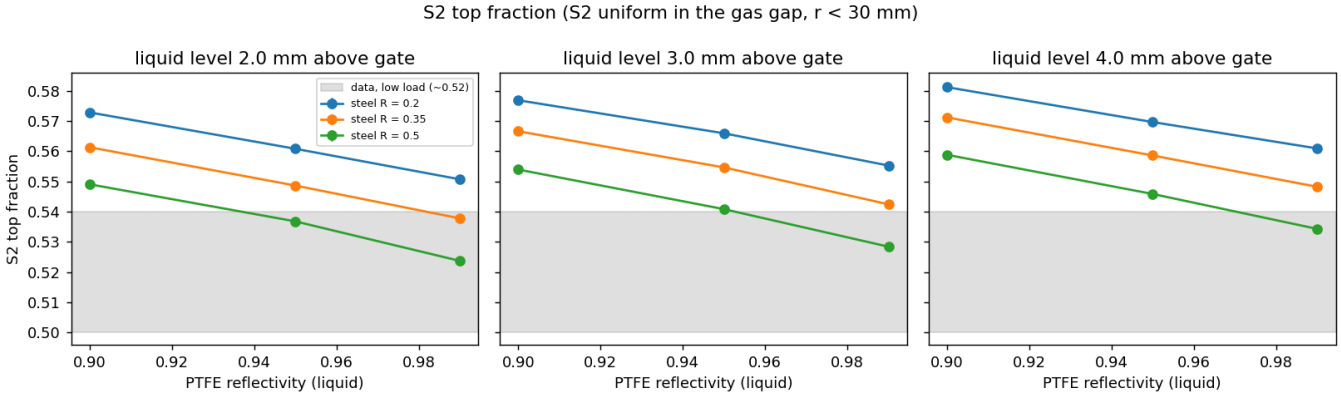
2. Validation

Each implementation step ended with a check against an analytic result:

check	result
solid angle (point source, 48.5 mm cathode at 10/30/60 mm)	detected fraction = solid angle, pulls ≤ 1.1
Fresnel + total internal reflection at the liquid surface, 0–80°	matches (Rs+Rp)/2; TIR from the critical angle 36.3°
mesh transparency, θ 0–75°, φ 0/29.4/45°	Geant4 = analytic ray trace photon by photon ($3 \cdot 10^6$ photons); 0.888 at normal incidence
optical surfaces (PTFE, steel)	reflected fraction = R, specular share = f, diffuse part Lambertian; R = 1 closes
sources and bookkeeping	z ranges follow the liquid level; every photon counted once
seed / statistics	independent halves of every scan point agree (worst pull 2.3)

3. Results

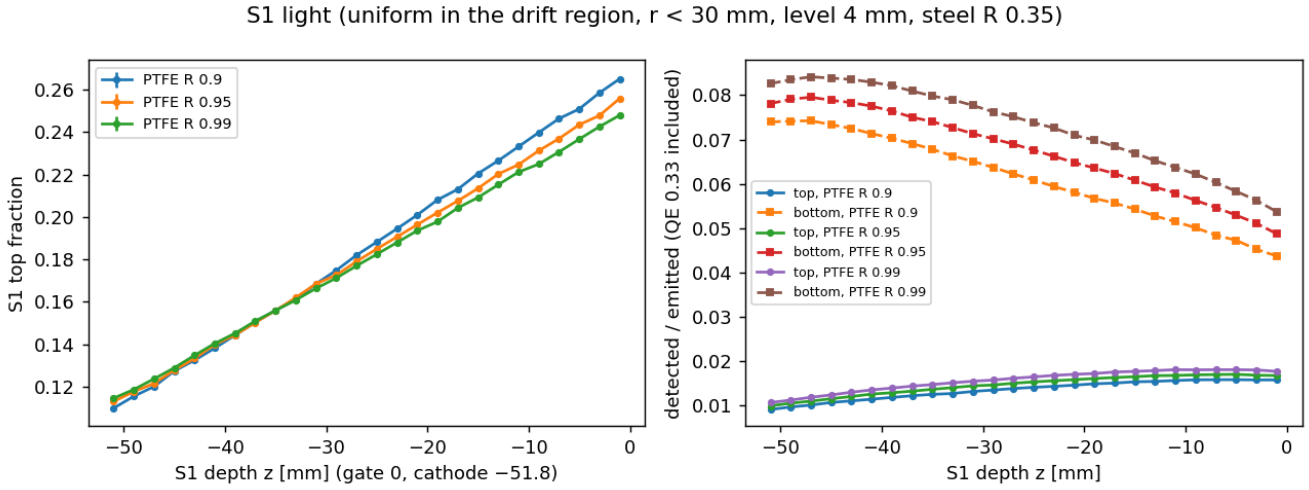
S2 top fraction. Over the full ranges of the free parameters and liquid levels: 0.524–0.584. Strongest knobs: PTFE specular fraction (−0.032 from 0 to 0.2), steel and PTFE reflectivity (−0.025 / −0.023 over their ranges), Rayleigh length (−0.012); liquid level 2 → 4 mm only +0.010. Versus radius 0.61 at the centre → 0.51 at the edge.



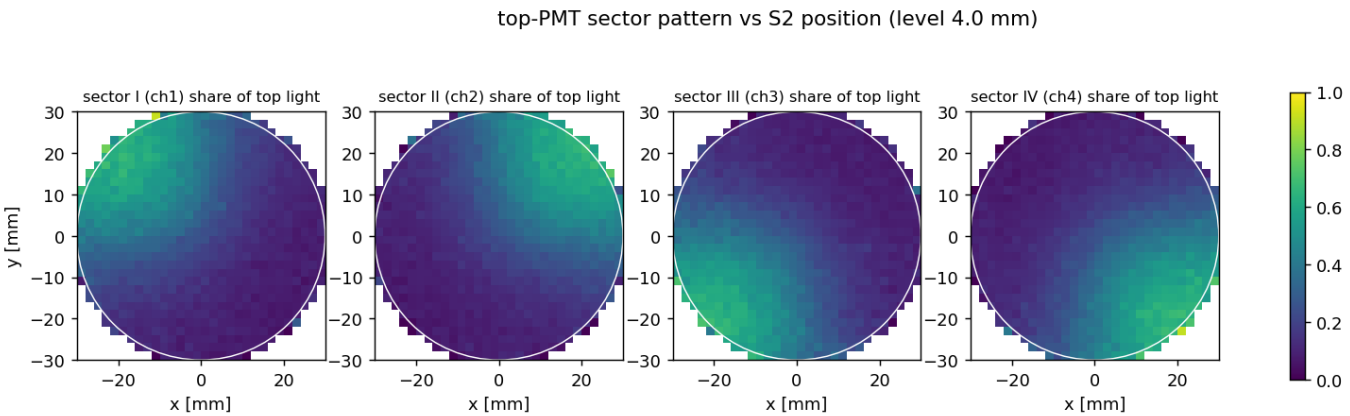
S2 top fraction vs PTFE reflectivity for three steel reflectivities and three liquid levels (10^7 photons per point; statistical errors ±0.0006).

Where the light ends. Reference point: level 4 mm, PTFE R 0.95, steel R 0.35, copper R 0.3.

	S2	S1
detected top / bottom PMT (QE incl.)	4.5 % / 3.6 %	1.5 % / 6.7 %
photocathode reached, not converted (1 − QE)	16.5 %	16.6 %
cryostat vessel (steel)	10.3 %	0.9 %
PTFE bucket	1.7 %	0.7 %
PTFE PMT holders top / bottom	2.0 % / 0.3 %	0.5 % / 0.6 %
mesh rings + foil rims	13.2 %	11.8 %
mesh foils: anode / gate / top screen / cathode / bottom screen	13.1 % / 10.2 % / 7.4 % / 4.5 % / 4.7 %	2.8 % / 16.3 % / 1.9 % / 11.7 % / 10.3 %
PTFE drift-region cylinder	2.8 %	9.4 %
PMT frame / body	2.5 %	3.5 %
copper field-shaping rings	1.7 %	2.0 %
absorbed in the liquid	0.9 %	2.4 %



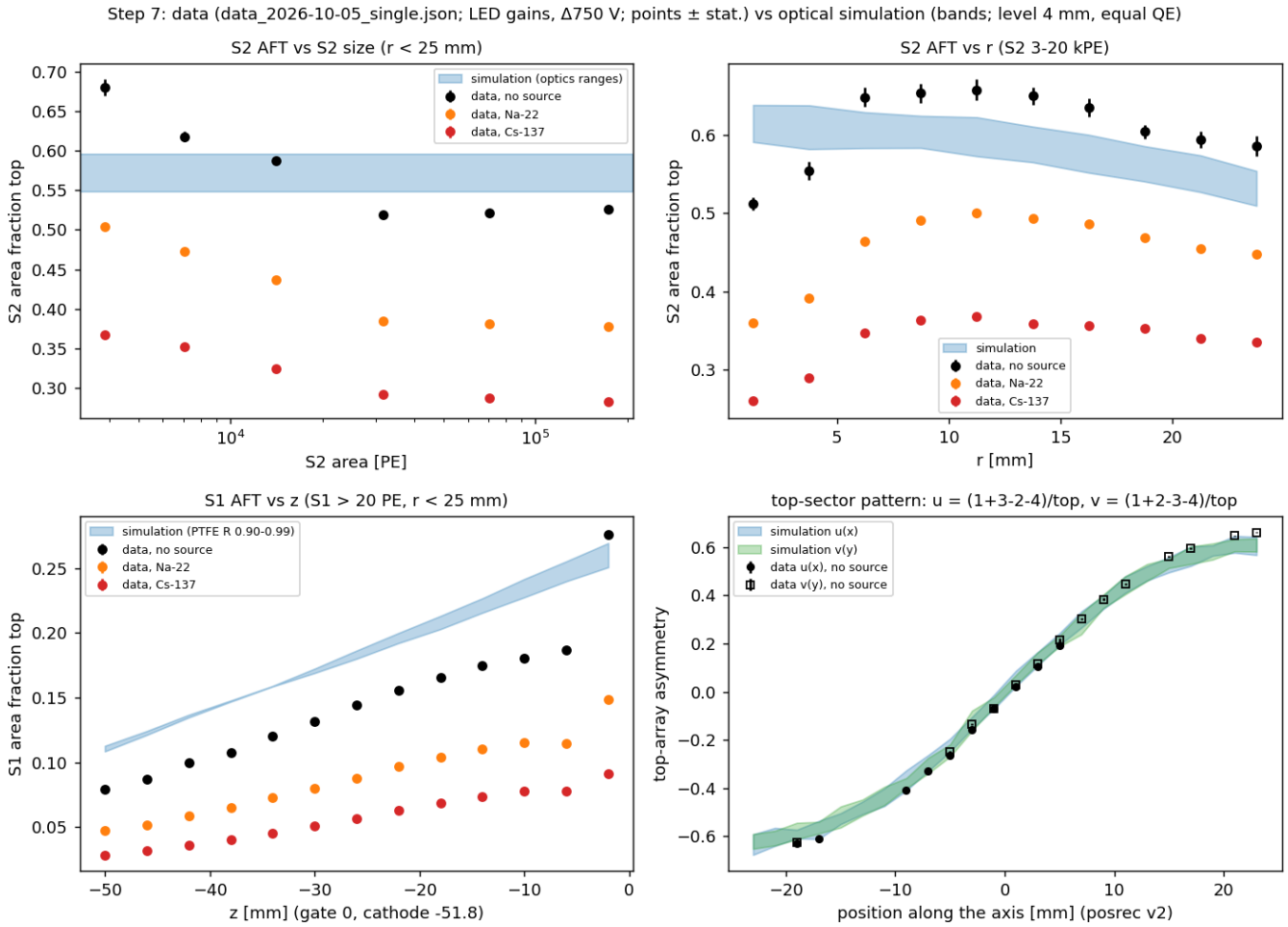
S1: top fraction (0.11 at the cathode → 0.25–0.27 at the gate) and light collection vs depth.



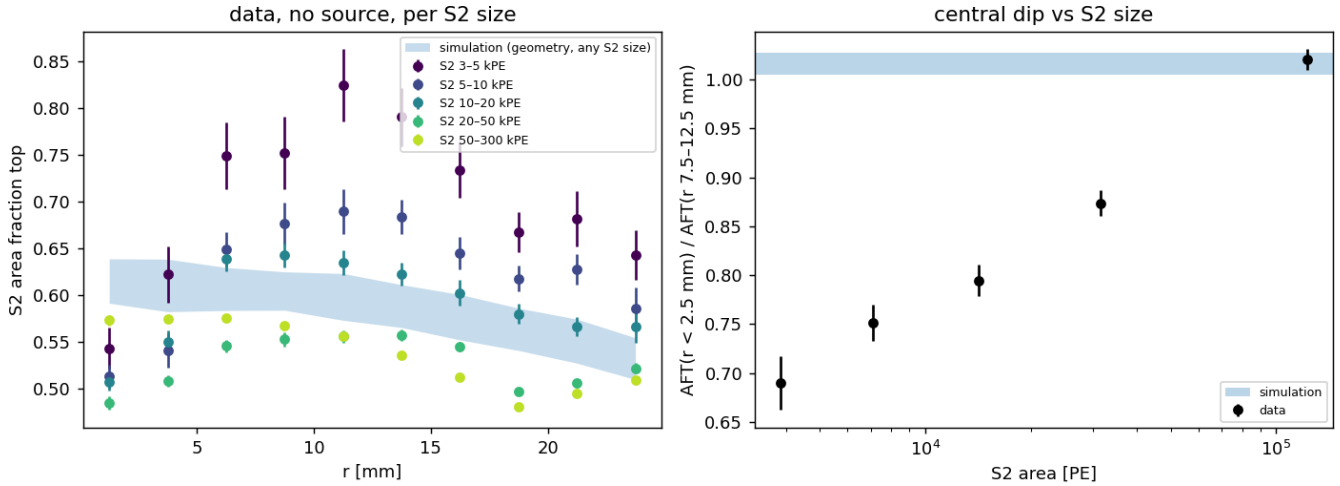
Share of each top-PMT sector in the top light versus S2 position (simulation frame: sector I at −x+y, II +x+y, III −x−y, IV +x−y).

4. Comparison with the data

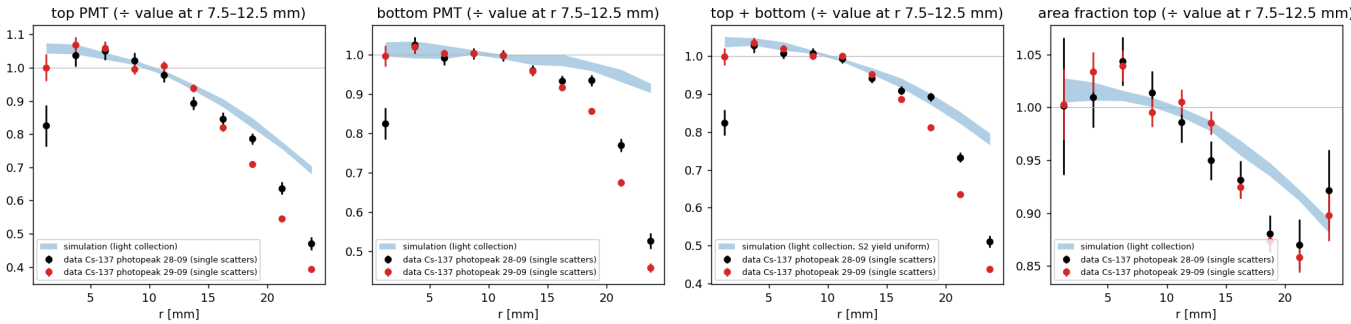
Data: Δ750 V runs of 28/29-09 (central processing v4), no source / Na-22 / Cs-137, **single scatters** (no alternative S2; 20–25 % of the events — a loose cut, alternative S2 < 5 % of the main one, gives the same), r < 25 mm, area fractions with the LED gains only (the v2 load correction f_top undone). Simulation band: all optics points at the 4 mm level, equal QE top and bottom.



Data (points, statistical errors) vs simulation (bands: all optics settings): S2 area fraction top vs S2 size and vs r, S1 vs z, and the top-sector asymmetries u(x), v(y).



Central dip of the S2 area fraction top versus S2 size (data, no source): per size class versus r (left), and centre / ring ratio (right).

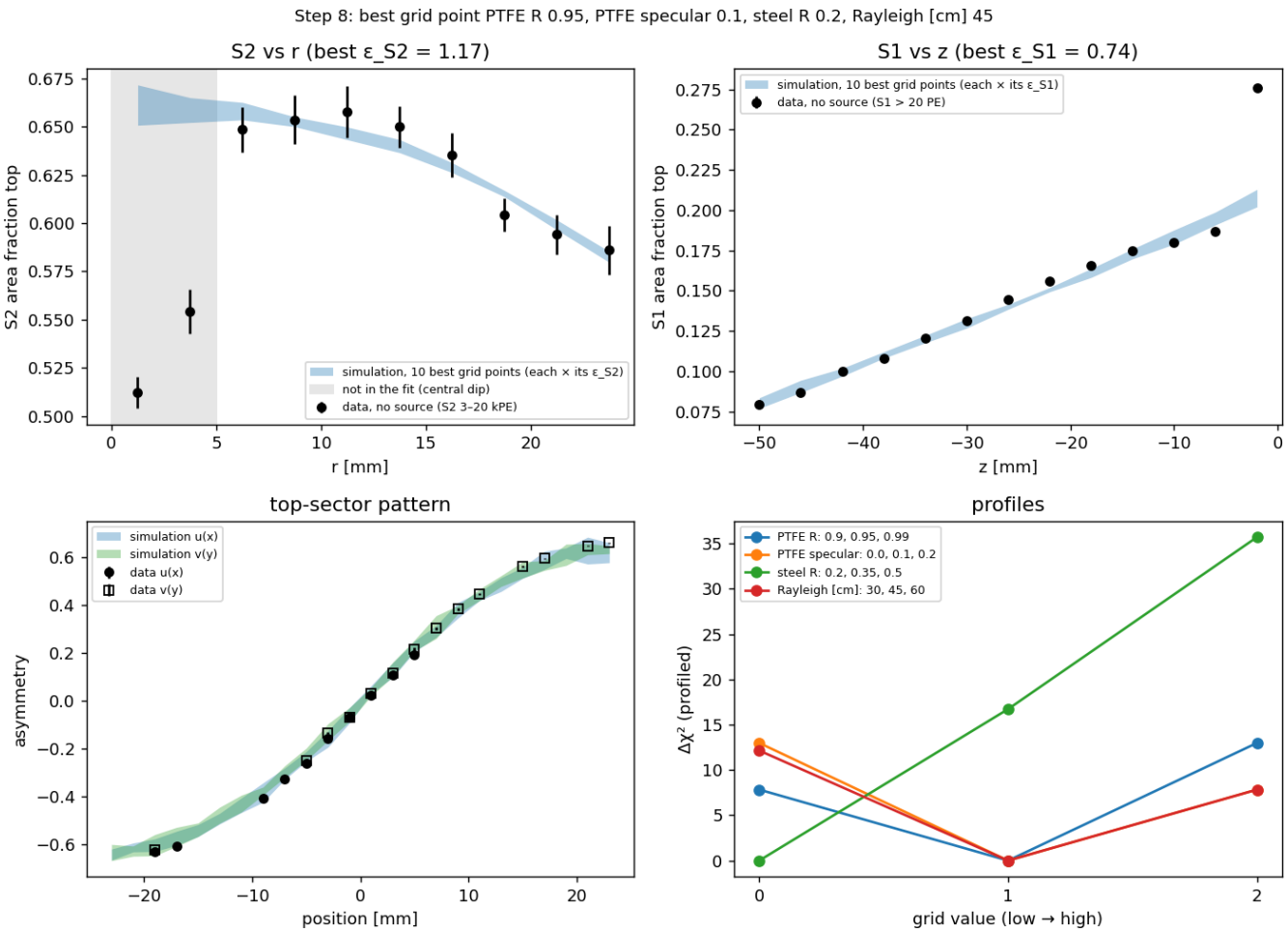


Top deficit or bottom excess? Cs-137 photopeak (fixed S2 per event, LED gains): top, bottom, total and area fraction top versus r, each ÷ its value at r = 7.5–12.5 mm; data points vs simulation band.

• **Sector pattern:** data inside the simulation band over ±24 mm, identical for all sources — geometry right.
• **S2 versus size and source:** no source 0.68 (3–5 kPE) → 0.53 (> 100 kPE); Na-22 0.50 → 0.38; Cs-137 0.37 → 0.28. Rate and size dependence are top-PMT effects (load, large pulses), not geometry. The simulation (0.55–0.60) agrees with the low-load, medium-size data.
• **S2 versus r:** outside the centre data/simulation top/bottom ratio 1.16–1.28 (absorbed by ε_{S2}); at r < 5 mm 0.67–0.79 (central dip in every source, also for single scatters).
• **S1 versus z:** same shape, ratio 0.68–0.76 at every depth (one efficiency factor for small signals on the top PMT).
• **Top or bottom?** Cs-137 photopeak (fixed S2 per event), single scatters: at the centre the area fraction top shows no dip (1.00 vs simulation 1.01–1.04) — the dip is a small-S2 effect. Outwards top and bottom both fall; the area fraction top at r = 24 mm is 0.90–0.92 (simulation 0.88–0.90), but the total S2 light drops to 0.44–0.51, far more than the optical ~0.78: charge or S2 light is lost near the wall. (With all events the bottom seemed to rise outwards — that was a multiple-scatter effect.)
• Implied top-PMT efficiency relative to geometry (S2): ≈ 1 without source, ≈ 0.3 for Cs-137 large S2 — the v2 gain factor f_top (0.74–0.88, anchored on Cs runs) corrects only part of it.

5. Fit of the optical parameters

81-point grid (PTFE reflectivity 0.90/0.95/0.99 × PTFE specular fraction 0/0.1/0.2 × steel reflectivity 0.2/0.35/0.5 × Rayleigh length 30/45/60 cm) for S2 and S1, binned like the data. Per point two efficiency factors multiply the top/bottom ratio (ε_{S2}, ε_{S1}); the optical parameters are constrained by the shapes (S2 vs r for r ≥ 5 mm, S1 vs z, u/v), with a systematic floor on each observable.



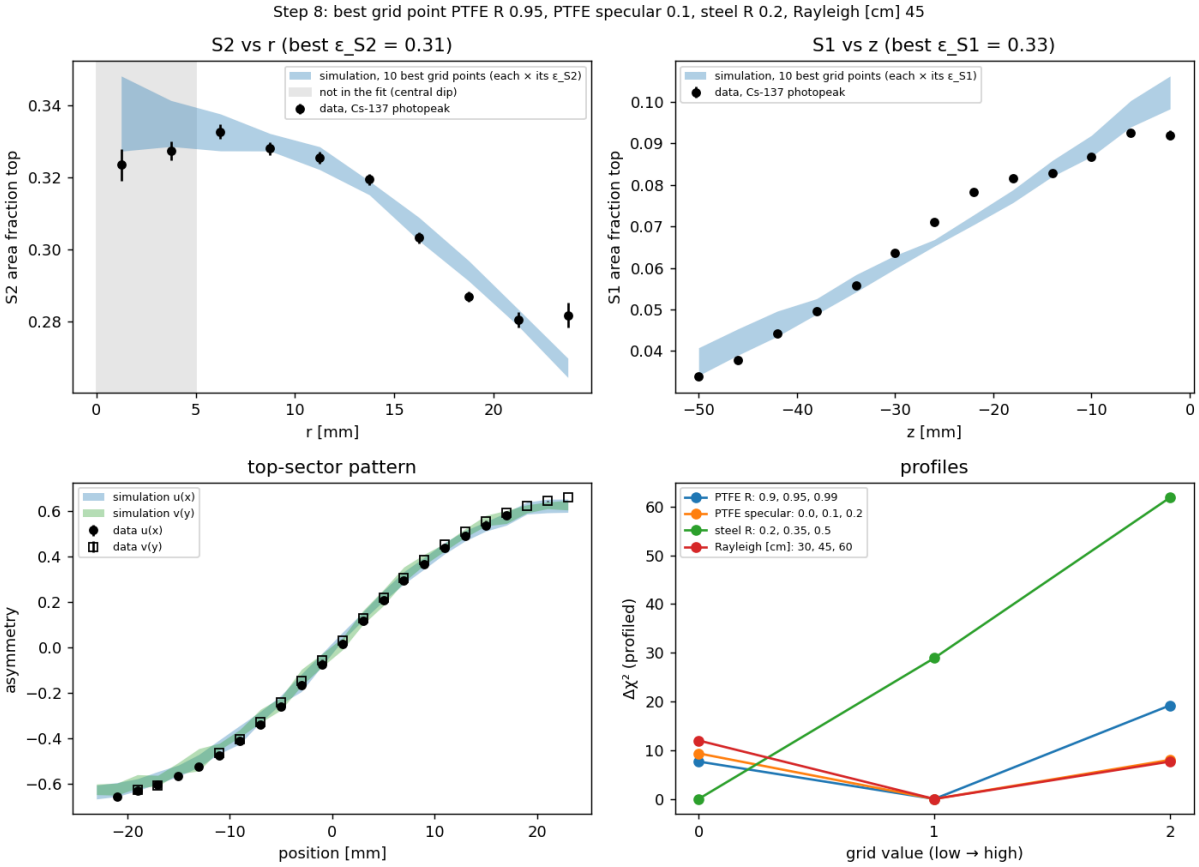
Fit: simulation band = the 10 best grid points (each scaled with its own ε), data = points (no source); χ² profiles of the four optical parameters.

• Best point: PTFE R 0.95, specular 0.1, steel R 0.2, Rayleigh 45 cm; χ² 2.0 / 8.0 / 27.4 for 7 / 11 / 24 dof (S2 vs r / S1 vs z / u, v).
• ε_{S2} = 1.17 (1.11–1.23), ε_{S1} = 0.74 (0.71–0.77): not one factor — the top PMT loses part of its small S1 signals (hit-only integration, section 7); for large signals (Cs-137 photopeak, section 6) both factors agree.
• Steel reflectivity low (0.2 preferred: Δχ² +16.7 for 0.35, +35.7 for 0.5). PTFE reflectivity, PTFE specular part and Rayleigh length are not constrained beyond the scan statistics: their Δχ² reach 8–13, but the best values moved between geometry v3 and v4 (PTFE R 0.99 → 0.95, Rayleigh 60 → 45 cm) while ε and the shapes stayed the same.
• Dead cross on the top PMT (0 / 1.5 / 3 / 6 mm): centre / ring 1.02 → 1.01, data 0.82 — not the cause of the central dip.
• Central dip versus S2 size (data, single scatters): centre / ring 0.69 (3–5 kPE) → 0.87 (20–50 kPE) → 1.02 (> 50 kPE) — explained in section 8 by an event population, not by the optics.

6. Large signals: the Cs-137 photopeak

The same fit with Cs-137 photopeak events only (Δ750 V, single scatters, cS1 photopeak ±0.5 σ, 34 600 events; S2 ≈ 120 kPE, S1 ≈ 1400 PE — no small signals).

	no source	Cs-137 photopeak
best optics	PTFE R 0.95, spec. 0.1, steel R 0.2, Rayleigh 45 cm	PTFE R 0.95, spec. 0.1, steel R 0.2, Rayleigh 45 cm
ε_{S2} / ε_{S1}	1.17 / 0.74	0.31 / 0.33
χ^2 S2 vs r / S1 vs z / u,v (7 / 11 / 24 dof)	2.0 / 8.0 / 27.4	3.3 / 6.4 / 43.9
AFT centre ÷ ring (data)	0.82	1.00

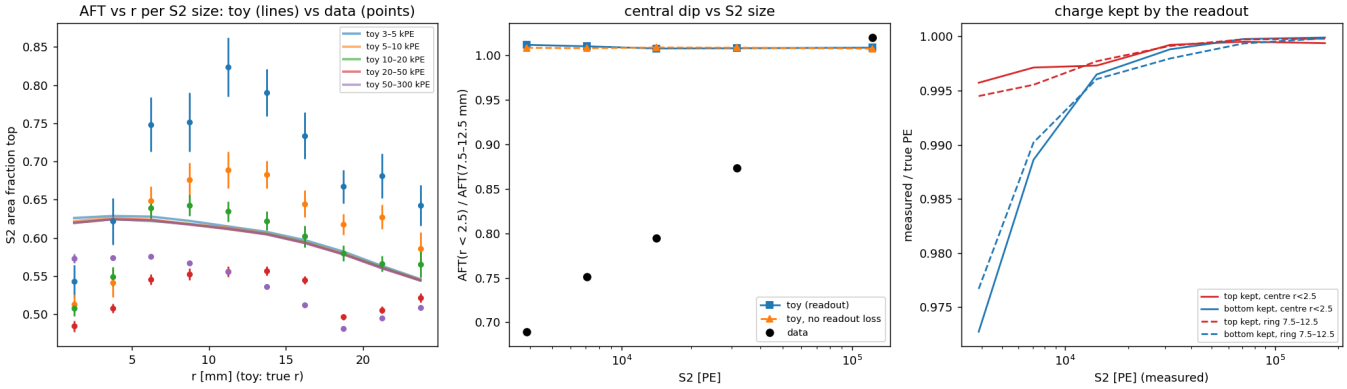


Fit with Cs-137 photopeak events: S2 vs r, S1 vs z, top-sector pattern, χ^2 profiles.

• Large signals: **one efficiency factor** for S1 and S2 (≈ 0.32, the top-PMT gain drop under Cs-137 load), the same optics as without source, no central dip. The optical model describes large signals.
• Best optics against the no-source data per S2 size (ε only): ε = 1.78 (3–5 kPE), 1.26, 1.07, 0.80, 0.78 (> 50 kPE): stable above 20 kPE, rising below (the falling AFT with size). Large S2s at y > 0: χ^2 10/7, centre ÷ model 1.02; at y < 0: χ^2 146/7 (section 8).

7. Readout: thresholds and charge sharing

Toy on top of the simulated light response: photons per channel and 4-ns sample from the S2 time profile, LED gains, measured single-photon pulse shapes, noise; V1725 self-trigger (top 20/30 ADC, bottom 60 ADC, pre-trigger 32 and 24 samples under threshold); amstrax hit finding. **amstrax sums only samples ≥ 15 ADC** (strax.sum_waveform on hits, no integration margin), so a lone top photon keeps 52–54 % (measured 55–60 %).

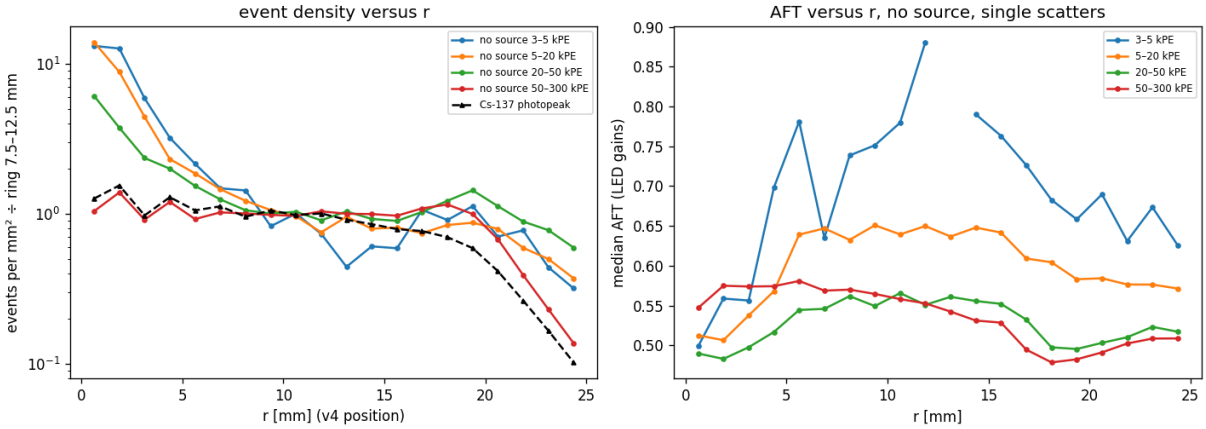


Toy vs data: AFT vs r per S2 size (left), centre/ring vs S2 size (middle), charge kept by the readout (right).

• S2 ≥ 3 kPE loses < 1 % (top) / < 3 % (bottom), centre and ring alike: the readout makes neither the central dip nor the AFT size dependence, and adds only 0.3 % to the Cs-137 photopeak S2 spread.
• S1: the top keeps 0.56 / 0.70 / 0.85 at 30 / 100 / 300 PE — the ε_{S1} of the no-source fit, and an S1 non-linearity. Bottom PMT at −800 V (≈ 3 ADC peak per photon, 60 ADC trigger): records nothing below ~1 kPE; the toy's bottom numbers for small signals need the real bottom threshold (to check). Fix: hit integration bounds and a lower hit threshold in amstrax (TODO).

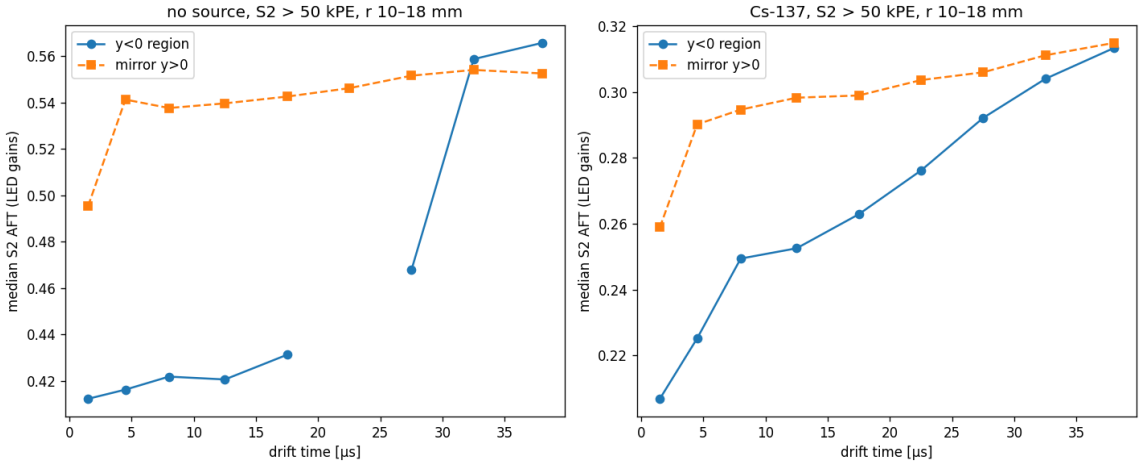
8. 2D maps: the central population and the y < 0 region

AFT and event density on x–y and on the raw top asymmetries u–v (no position reconstruction), per S2 size, no-source and Cs-137 data (Δ750 V, single scatters, LED gains), against the simulated map.



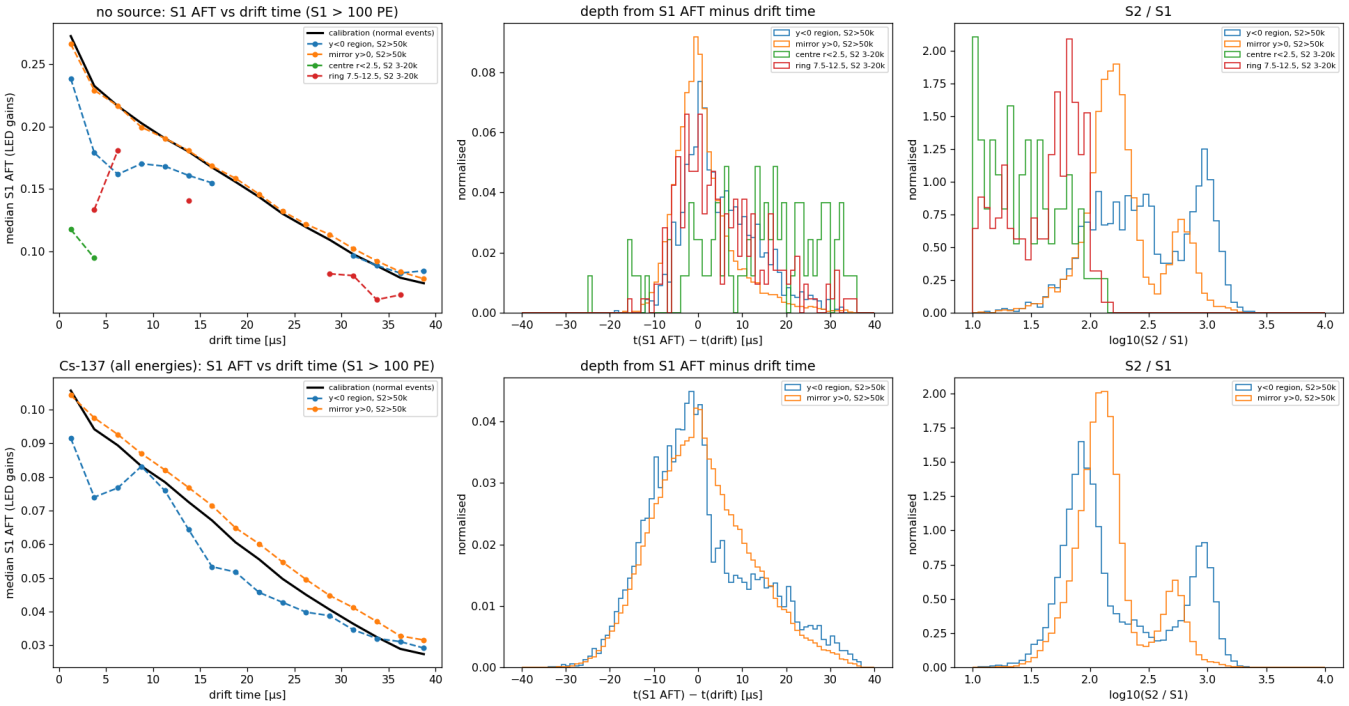
Event density (÷ ring 7.5–12.5 mm) and AFT versus r per S2 size class, no source.

• **Central population:** small no-source S2s pile up at the reconstructed centre (×14 / ×10 / ×4 / ×1.2 for 3–5 / 5–20 / 20–50 / > 50 kPE; Cs photopeak flat). They spread their top light equally over the four anodes (u ≈ v ≈ 0) and have AFT ≈ 0.5: the central dip is this population. Their S1 does not belong to the drift time (S1 AFT vs drift time below: 70 % off by > 10 µs, S1 15 µs deeper; S2/S1 ≈ 13 vs 34 in the ring): likely accidental pairs of an S1-only event (e.g. below the cathode) with a small diffuse S2 (delayed / single-electron trains spread over the TPC → u ≈ v ≈ 0).
• **Not a channel problem:** fitted relative top-channel gains 0.95–1.04, the same from Cs-137 and no source; the deficit is the same on the slow ch4 side and the fast ch3 side and the top-channel shares stay normal; position reconstruction anchored on the gate mesh would show a ch3/ch4 error of the needed size.
• **Cs-137 S2 yield:** top and bottom move together with position (±15–20 %); the top/bottom ratio is flat within ~5 % except at the edges. An x–y correction of the photopeak cS2 (cross-validated) reduces the spread 47 → 41 % at r < 25 mm, 42 → 40 % at r < 20 mm and not at all at r < 15 mm: not the cause of the ~40 % S2 spread.



AFT versus drift time in the y < 0 region (r 10–18 mm) and its mirror at y > 0, S2 > 50 kPE, no source and Cs-137.

• **y < 0 region:** in an arc at y < 0, r ≈ 10–18 mm (also in raw u–v, next to cells with almost no events) the top/bottom ratio drops to ~0.6–0.8 × the model — only for interactions with drift time < 25 µs (AFT 0.41–0.43 vs 0.54 in the mirror region), not for deeper ones (0.56 vs 0.55); same on 28-09 and 29-09 and for Cs-137. Events there are shallower (median drift 4.3 vs 10.8 µs), have wider S2s (width50 739 vs 624 ns) and smaller S1s. Cs-137 **photopeak** events in the region show it only mildly (AFT ×0.88–0.95 at drift < 20 µs): the strong deficit belongs to events with an S1 too small for their S2.
• **S1 AFT versus drift time** (figure below): S2/S1 has two populations everywhere — bulk events (≈ 100–200) and near-gate events (≈ 600–1000, drift ≈ 2 µs, S1 AFT 0.24–0.27 = consistent with ~2 mm below the gate). Near-gate S2s have a lower AFT everywhere (−0.04 to −0.05). In the y < 0 region they are 52 % (no source) / 41 % (Cs) of the events vs 22 % / 17 % in the mirror, mainly because bulk background events concentrate at y > 0. Cs-137 bulk events: region 0.293 vs mirror 0.300 — **normal events at y < 0 agree with the model**. Open: why near-gate S2s have a lower AFT, and a residual for near-gate events at y < 0 (0.41 vs 0.50).



S1 AFT versus drift time (left), depth from the S1 AFT minus drift time (middle) and S2/S1 (right) for the y < 0 region, its mirror, the central small-S2 population and its ring; no source (top) and Cs-137 (bottom).

9. Design study: extra PTFE cylinders

PTFE cylinders with the radii of the drift-region PTFE, inside the ring bores, foil to foil:

extra PTFE	S2 light	S2 vs now	S1 light	S1 vs now	S2 top fraction
none (as built)	8.1 %	+0 %	8.1 %	+0 %	0.559
gate–anode	9.6 %	+18 %	8.8 %	+9 %	0.549
anode–top screen	9.3 %	+15 %	8.3 %	+3 %	0.585
cathode–bottom screen	8.5 %	+5 %	9.2 %	+14 %	0.537
gate–anode + anode–top screen	11.0 %	+36 %	9.1 %	+12 %	0.573
gate–anode + anode–top screen + cathode–bottom screen	11.6 %	+43 %	10.4 %	+28 %	0.550

Most rescued photons still end on meshes, PMT frames or in the photocathode (QE), so the gain is a factor ~1.4. Not studied: high voltage (PTFE across 5 kV gate–anode, charge-up in the extraction region), mechanics.

10. Anode mesh rotated by 120°

A square mesh repeats every 90°, so a 120° rotation of the anode relative to the gate is a 30° rotation. Simulated with the anode as now (aligned with the gate) and rotated ($4 \cdot 10^7$ photons each):

	anode as now	anode +120°
S2 top fraction	0.5595 ± 0.0003	0.5599 ± 0.0003
S2 light collection top / bottom	4.53 / 3.57 %	4.54 / 3.57 %
S2 light absorbed on the anode foil	13.0 %	13.0 %
S1 top fraction	0.1791	0.1792
fine structure of the top light vs S2 position (1 mm, beyond statistics)	3.0 %	3.0 %

Optically nothing changes (radial and depth profiles and the sector pattern coincide as well). Not covered: the electric field — a rotation changes where gate and anode bars cross, and with it the local extraction and electroluminescence field.

11. Limitations and next steps

• Surroundings are in (v4: PTFE bucket, cryostat vessel, PMT holders); ~10 % of the S2 light ends on the vessel. Still missing: the three gate–anode pillars; bucket bottom, vessel ends and the gap below the bucket are defaults.
• Central S2 dip: identify the small-S2 population at the centre (S2 width, drift time, S1, S1-AFT vs drift-time z).
• Near-gate events: why their S2 AFT is lower (S1 light on the top PMT ~2 μs before the S2? extraction near the gate wires?); the central population: confirm accidental S1–S2 pairing (S1 width / pulse shape, S2 width, time since the previous large S2).
• Optical parameters: a finer grid with more photons (u/v term) to constrain PTFE reflectivity and Rayleigh length. amstrax: integration bounds + lower hit threshold so small signals keep their full charge.
• Oct 2024 (2 + 6 mm liquid/gas) vs now (4 + 4 mm): the simulation predicts only +0.01 in the S2 top fraction.